

P. SHUBHANG SHARMA

PROFILE	Trained in mathematics and physics, with a strong interest in expressing abstract scientific ideas through more human ways of writing, visual media and talks. My current independent break centres on preparing for a research career in the intersections of mathematics and physics while retaining the values of art, science and communication intact.	
EDUCATION	National Institute of Science Education and Research (NISER) ↗	2020-2025
	<i>Integrated M.Sc. in Physics Bhubaneswar, India</i>	CGPA: 8.69/10
	FIITJEE Junior College ↗	2018-2020
	<i>Higher Secondary Education Hyderabad, India</i>	Score: 98.2%
	Top 1% in State (Telangana, India).	
LANGUAGES	English, Hindi, Telugu (fluent) and Kannada (conversational)	
PUBLIC ENGAGEMENT	Undergraduate Committee of the Institute, NISER	2022-2023
	<i>Student representative</i>	
	· Represented the School of Physical Sciences in institute-level academic discussions, facilitating dialogue between students and faculty, over curriculum, mental health, learning spaces etc.	
	NISER Film Club's Interview Team	Spring 2023
	<i>Co-producer, editor and interviewer</i>	
	· Conducted an interview with a faculty member reflecting on scientific, institutional, and personal experiences within academic life.	
WRITING AND PUBLIC SCHOLARSHIP	The Ideas of Real Analysis	Fall 2025 - present
	<i>PDF Version</i>	
	<i>Part of an ongoing experimental series to express abstract mathematical ideas via more human narratives.</i>	
	On the Geometry of Curves and Surfaces	Spring 2024
	<i>Narrating the key highlights of a course on curves and surfaces.</i>	
	Hydrogen Atom: The final take?	Fall 2023
	<i>Revisiting hydrogen atom, with all its fine structures.</i>	
	An introduction to Mathematical Physics	In progress
	<i>A guiding perspective on its history and contemporary ideas, mostly through a geometric lens.</i>	
	The Ideas of Quantum Field Theory	In progress
	<i>PDF Version</i>	
	<i>On how QFT explains fundamental interactions in nature and a presentation of some standard calculations in the subject.</i>	
AWARDS /SCHOLARSHIP	DST-INSPIRE Scholar ↗	2020 – 2025
	<i>National Scholarship for Higher Education (SHE) awarded for ranking in the top 1% of performers in Higher Secondary Education</i>	
HOBBIES	Running, cycling, hiking, badminton, music, origami, cooking, blogging.	

CREATIVE TECHNOLOGY

Creative Coding

Art × Code × Science

[Gallery](#)

[Explore](#)

- Experience with the JavaScript library **p5.js** for artistic, interactive, physics-based visualizations.

Web Development (HTML, CSS, JavaScript)

[Personal webpage](#)

Communication × Code × Art

- Experience with HTML, CSS and JavaScript to build minimal websites for scientific and personal communication. Interested in exploring web development as an interactive story-telling medium in the space of public science.

Vector Illustrations (INKSCAPE)

[Gallery](#)

- *Some mathematical expressions using vector art.*

Computational Physics

[Library](#)

Int. MSc. Coursework

- Built a numerical-methods Python library with ODE/PDE solvers along with numerical integrators, fitting, root-solving algorithms.

MASTER'S DISSERTATION

[AdS/CFT Correspondence and Conformal Geometry](#)

Fall 2024, Spring 2025

Advisor: Dr. Yogesh K. Srivastava

Studied the foundations of the AdS/CFT correspondence, a framework relating gravity and quantum field theory through boundary–bulk interactions, and its deeper connections to geometry.

GRADUATE COURSEWORK

Introduction to Manifolds. Quantum Field Theory I & II, General Relativity, Particle Physics, Astronomy and Astrophysics.

ACADEMIC PROJECTS

[Notes on Differential Geometry \(*in progress*\)](#)

Spring & Fall 2025

Course: Introduction to Manifolds, Instructor: Dr. Chitrabhanu Chowdhury

Consolidated lecture notes on the course material involving the generalization of Euclidean spaces to higher dimensions having more abstract and interesting properties.

[Semester Reading project on Gauge Theory](#)

Spring 2024

Course: Quantum Field Theory II, Instructor: Dr. Yogesh K. Srivastava

Studied how geometry plays a fundamental role in governing the interactions of the force particles in nature. Occurs as a fertile ground for a two way development between math and physics.

[Open Lab experiment on Energy Transfer in Electric Circuits](#)

Fall 2023

Course: Integrated Physics Laboratory I, Instructor: Dr. Kartikeswar Senapati

Built a 39m long transmission-line model to demonstrate the artefacts of a field-based energy transfer by recording nanosecond level voltage steps across the load.

[Term Paper on Finite Difference Methods](#)

Spring 2023

Course: Computational Physics, Instructor: Dr. Subhasish Basak

Implemented finite difference methods to solve PDEs with different levels of improvements. Demonstrated these methods on the omnipresent Laplace equation, which for example governs the shape of a thin and tightly stretched rubber sheet.